

**M from Mystery:
Exceptional Kernels, Global Charge Data, and the Electron
as Atomos**

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Abstract

What mathematical statement would mean that the electron is indivisible? The experimental statement, that nothing has been found inside it, is bounded by resolution. The representation-theoretic statement, that it carries an irreducible label, is refuted as evidence by the proton, which occupies an irreducible spin-half sector while being made of quarks. The field-theoretic statement, that it appears as an elementary field, is presentation-dependent, since exact dualities exchange elementary fields with composites. And the object itself outruns its label: the physical electron is an infrared-dressed sector that no bare Wigner particle describes. This paper organizes those four obstructions into a research programme. Its established floor is new: in 2026, the exact Standard Model gauge group with its global Z_6 quotient was derived as a stabilizer inside the automorphisms of the exceptional Jordan algebra, the full chiral content of one generation was recovered from Peirce decompositions of a Jordan pair, and e_7 was shown to contain three generation-shaped spaces. The paper defines the structure those results hint at: a flagged Jordan-spectral kernel and its quantum completion M , quotiented by renormalization and exact duality, with emergence stated as a classification problem whose axioms may not hide the numbers 3, 2, 1, or 16. Everything on the floor is then computed exactly. The 16-state generation cancels all 5 consistency conditions in exact rational arithmetic, and deleting any multiplet breaks at least one, except the right-handed neutrino, which breaks none: the one particle the kernel adds beyond observation is the one the ledger does not demand. The 4 global forms of the gauge group are built as explicit line lattices: identical matter, 4 distinct theories, so the Jordan construction's selection of the Z_6 form is a physical claim, and its Langlands shadow is a computable dual lattice rather than an analogy. The $SL(2, \mathbb{Z})$ orbit of the unit electric charge is verified to be the coprime vectors, at density 0.598 against the asymptotic $6/\pi^2$, and every primitive vector decomposes into nonzero constituents: primitive labels never establish indivisible carriers. In the fusion ring of $SU(2)$, every irreducible is tensor-prime and every spin above one half is a summand of a product of smaller spins, so the bound-state relation is a summand and never a factor, and primality with respect to the tensor product detects nothing. These computed no-gos force a 7-level hierarchy of atomicity ending in duality-atomic and UV-atomic, define the electron object as a dressed cone over the electroweak vacuum, and state the electron conjecture with its final loophole visible: primality is relative to the theory in which composition is defined, and even the strongest theorem would prove indivisibility within M , never beneath it.

The question

Democritus named the atom for what could not be cut. Physics has kept the word and lost the criterion, because every operational meaning of "cannot be cut" has failed in an instructive way. This paper is about what a defensible criterion would look like, how far the mathematics of 2026 actually goes toward one, and what remains conjecture. Four obstructions define the problem.

Finite resolution first. No experiment establishes absence of structure at all smaller scales; the electron is pointlike down to the shortest distances probed, which is a statement about the probes. Second, the proton. As a one-particle state it furnishes a perfectly irreducible representation of the

Poincaré group, mass m_p and spin one half, in the classification where a particle is an irreducible unitary representation labeled by mass and spin (Wigner, 1939). It is also three quarks and glue. An irreducible label therefore cannot certify a noncomposite carrier, and any argument from the electron's labels to its indivisibility dies against this example. Third, duality. In supersymmetric gauge theories there are exact equivalences under which a composite operator of one description is an elementary field of the dual description (Seiberg, 1995; Fang, Feng and Xie, 2024), and the oldest form of the phenomenon, electric-magnetic duality, exchanges the elementary electric quanta with solitonic monopoles (Montonen and Olive, 1977). "Elementary field in this Lagrangian" is a fact about the Lagrangian. Fourth, dressing. A charged particle in quantum electrodynamics cannot be separated from its arbitrarily soft photon cloud; the physical charged state is an infraparticle with no sharp mass shell, a theorem rather than a technicality (Fröhlich, Morchio and Strocchi, 1979; Buchholz, 1986; Duch and Dybalski, 2023). Whatever the candidate atomos is, it is the dressed sector, never the bare field.

Each obstruction removes a cheap answer, and together they set the specification. A meaningful indivisibility claim must be about an object rather than a label, in a category rather than a Lagrangian, invariant under every exact duality, and stated for the dressed sector. The framework that already meets the first two conditions is algebraic: superselection sectors form a tensor category whose product encodes how charges compose (Doplicher, Haag and Roberts, 1971), and from that category the global gauge group can be reconstructed (Doplicher and Roberts, 1989). What no existing framework supplies is the third and fourth conditions together with a structure in which the Standard Model itself is output rather than input. Call the conjectural structure M . The name is honest: no single presentation reveals such an object, and we would observe only its shadows, spectra, observables, defects, dualities, and low-energy phases.

The established floor

The reason this programme is worth stating now is that its kinematic floor stopped being hypothetical in the summer of 2026. Three results, presented here as their authors present them.

The gauge group. Baez and Schwahn proved that the subgroup of the automorphism group of the exceptional Jordan algebra $h_3(O)$ preserving a nested pair of subalgebras is isomorphic to $S(U(2) \times U(3))$, which is precisely $[SU(3) \times SU(2) \times U(1)]/Z_6$, the Standard Model gauge group in its exact global form, Z_6 quotient included (Baez and Schwahn, 2026). The algebra itself is the one exceptional object in the 1934 classification of Jordan, von Neumann and Wigner, found in a failed algebraic reformulation of quantum mechanics and unemployed for ninety-two years (Jordan, von Neumann and Wigner, 1934).

The matter. Baez, Bokor and Boyle showed that a chain of Peirce decompositions along a compatible flag of tripotents, running from $h_3(O_C)$ through the bi-Cayley space to $M_{\{3,2\}}(C)$, produces a 16-complex-dimensional space that decomposes under the residual group into the six multiplets of one Standard Model generation in left-handed Weyl form, hypercharges correct, right-handed neutrino included (Baez, Bokor and Boyle, 2026). The authors state plainly that the tripotents are chosen rather than derived, and that the construction's threefold triality structure has not been

turned into three generations.

The families. Baez showed that e_7 decomposes into a subalgebra plus three 32-dimensional spaces each shaped like a generation, and presents the result as representation theory with a chosen embedding, explicitly declining to call it a physical theory (Baez, 2026). The lineage behind all three results runs from octonionic quark structure (Günaydin and Gürsey, 1973) through the exceptional survey (Baez, 2002) to exceptional quantum geometry (Dubois-Violette, 2016; Dubois-Violette and Todorov, 2019).

The discipline this paper adds is a strict accounting of registers. Established: the exact global gauge group as a stabilizer; one generation's representation content from a chosen flag; three generation-shaped spaces in e_7 . Absent: any dynamics selecting the flag; the Higgs and Yukawa sector; three physical generations; a quantum theory. The distance between those two lists is the programme.

The kernel and its completion

The proposal, stated as architecture and marked as such. The finite kernel is a flagged exceptional Jordan structure with spectral data: the algebra, a real form, its cubic norm, a compatible tripotent flag, and a finite Hilbert space with charge conjugation, chirality, and a finite Dirac operator, the last carrying whatever masses and Yukawa couplings the kernel can express. Spectral geometry is the template: in the noncommutative construction of the Standard Model, the gauge and Higgs fields arise as inner fluctuations of a finite Dirac operator and the action from a spectral principle (Chamseddine and Connes, 1997), with the honest caveat that generations and Yukawa numbers are inserted there rather than derived (Devastato, Kurkov and Lizzi, 2019); the extension of that machinery toward exceptional and Jordan settings is under active construction (Farnsworth, 2025). The kernel proposal reverses the usual direction: the Jordan flag should generate the spectral data rather than the Standard Model algebra being written down first.

The completion M is the quantum object the kernel would generate: observables assigned to regions in the factorization-algebra sense (Costello and Gwilliam, 2023), sectors and defects, anomaly data with a chosen trivialization (García-Etxebarria and Montero, 2018), renormalization flow, and then a quotient by every exact duality, so that two Lagrangians presenting the same observables, sectors, defects, anomalies, and flow are the same point of M . The quotient is the operative step: it makes “elementary in presentation P ” inexpressible as a property of M , and that inexpressibility is why M is worth defining.

Emergence gets a definition rather than an adjective. Admissible kernels are those satisfying structural axioms, positivity, chirality, anomaly trivializability, a stable symmetry-breaking vacuum, no massless exotics, and the axioms are forbidden from containing the numbers 3, 2, 1, or 16 in disguise. The reconstruction ladder has rungs: the gauge Lie algebra, the exact global group, the chiral matter, the classical action, the quantum theory, the unique ultraviolet completion, and last the duality-invariant atomicity of the electron. The 2026 results stand on the second and third rungs. The central classification conjecture, that the exceptional kernel is an isolated admissible point, would convert “the Standard Model fits beautifully into exceptional mathematics” into “ev-

ery admissible structure is the Standard Model,” which is what deriving would mean. Nothing of the sort is proved, and the paper’s job is to make the statement precise enough to fail.

The one-generation ledger, computed

Everything on the floor can be computed, and this paper computes it exactly, in integer and rational arithmetic with no numerical tolerance anywhere. The first computation is the generation itself. Sixteen left-handed Weyl states in six multiplets, hypercharges one sixth, one third, minus two thirds, minus one half, one, and zero. Five consistency conditions: the two mixed gauge anomalies, the cubic hypercharge anomaly, the mixed gravitational anomaly, and the Witten parity requiring an even number of SU(2) doublets. All five vanish identically, as exact fractions summing to zero, with 4 doublets counted (Figure 1, left).

The sharper computed fact is rigidity. Delete any single multiplet and recount: removing the quark doublet breaks 5 conditions, the lepton doublet 4, the up conjugate and down conjugate 3 each, the electron conjugate 2. One deletion breaks nothing: the right-handed neutrino, singlet under everything with hypercharge zero (Figure 1, right). The generation is anomaly-rigid at every site but one, and the one exception is the one particle whose existence remains experimentally indirect. The construction that recovers the generation from Peirce decompositions delivers this multiplet as part of the package; the ledger explains why physics could afford to be undecided about it for so long. A structure that produces the 16-state pattern is therefore reproducing something far more constrained than a list: it is landing on the near-unique solution of a system of Diophantine conditions, and the near-uniqueness is measurable, one free site in six.

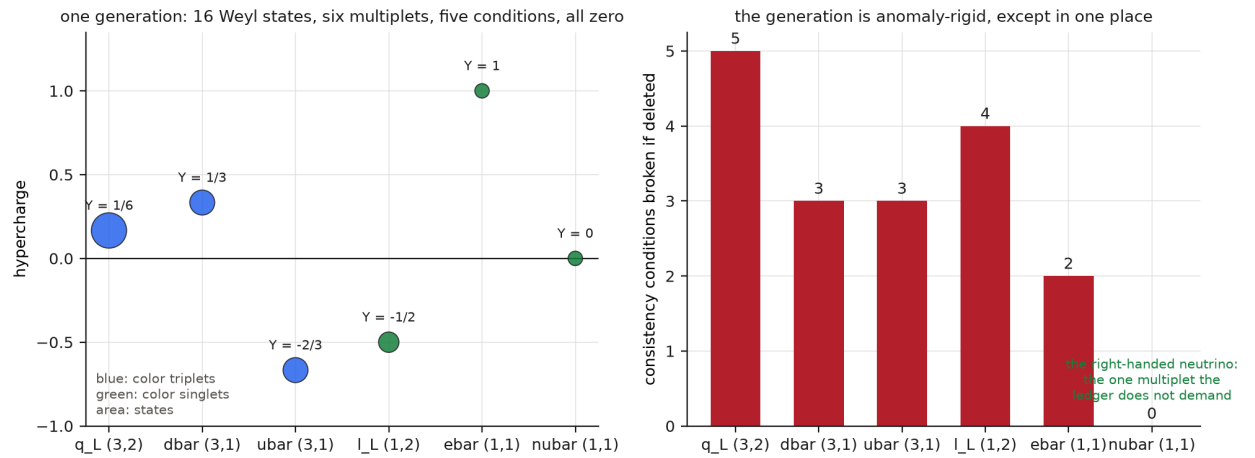


Figure 1: The one-generation ledger. Left: 16 Weyl states in six multiplets with exact hypercharges; all five consistency conditions vanish in exact arithmetic. Right: deletion rigidity: every multiplet’s removal breaks conditions except the right-handed neutrino, the one particle the ledger does not demand.

Four theories with identical matter

The gauge Lie algebra does not determine the theory. The center classes of $SU(3) \times SU(2) \times U(1)$ form a group $Z_3 \times Z_2 \times Z_6$, and the six matter multiplets occupy classes that generate a Z_6 inside it, the computation confirming that all six lie on the single diagonal generated by the quark doublet's class. Consequently there are four global forms, the quotients by the subgroups of Z_6 , and the paper builds all four as explicit charge lattices. Every multiplet is admissible under every form: perturbation theory, particle content, and every scattering amplitude built on them are identical across the four. What differs is the extended operator content. The allowed Wilson and 't Hooft line classes form lattices of sizes 6 by 1, 3 by 2, 2 by 3, and 1 by 6, each with the Dirac pairing verified integral, and the four lattices are pairwise distinct (Figure 2). Four theories, one matter content, distinguishable only by which probes exist, which is the modern statement of Tong's observation that experiment has not yet chosen among them (Tong, 2017).

This is where the Jordan results acquire physical rather than aesthetic content, and where the Langlands connection stops being decorative. The stabilizer theorem does not produce a Lie algebra; it produces $S(U(2) \times U(3))$, the full Z_6 quotient, which is one of the four lattices and one of the four theories. A structure that selects the global form is making a claim about which magnetic monopoles exist, and the claim has a dual side: the magnetic lattice of each form is the electric lattice of its Goddard-Nuyts-Olive dual group (Goddard, Nuyts and Olive, 1977), the object that number theory independently named the Langlands dual. The rigorous bridge between S-duality and the Langlands correspondence lives in twisted supersymmetric Yang-Mills, where Wilson and 't Hooft operators exchange under the dual group (Kapustin and Witten, 2007), with the modern spectral reading of the correspondence describing decomposition into atomic, monochromatic pieces (Ben-Zvi, 2026) and its boundary form pairing dual Hamiltonian spaces (Ben-Zvi, Sakellariadis and Venkatesh, 2024). None of that machinery applies to the Standard Model as it stands, and this paper follows the sources in saying so. The one disciplined conjecture worth stating is small and concrete: that the kernel's selected global form, with its computed electric and magnetic lattices, is the invariant a protected sector of any completion M must present to its dual description. The Langlands shadow of the electron, today, is a 6-by-6 grid of allowed charges, computed in this paper's simulation, and nothing grander.

Three shortcuts, three computed failures

The programme's endpoint concerns indivisibility, so the paper computes the failure of every label-based shortcut before defining anything stronger.

First shortcut: primitive charge. In the abelian theory with electric and magnetic charges, the duality group $SL(2, Z)$ acts on charge vectors, and the orbit of the unit electric charge consists of the primitive vectors, those with coprime entries, by Bézout's identity. The simulation verifies this constructively over the box of charges up to 40, building an explicit determinant-one witness matrix for each of the 3,920 primitive vectors among 6,560, a density of 0.598 approaching the asymptotic 6 over pi squared, which is 0.6079 (Figure 3, left). So "the electron's charge is a primitive lattice

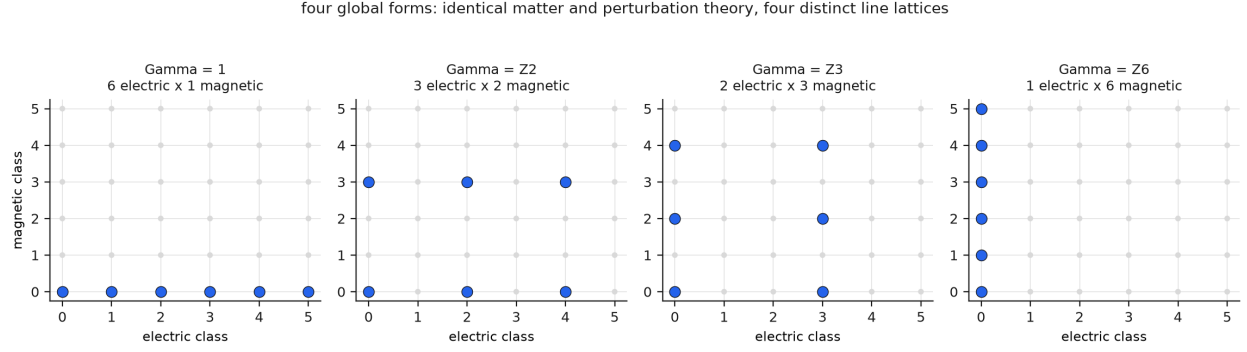


Figure 2: Four global forms of the Standard Model gauge group. Identical matter and perturbation theory across all four; the line lattices differ, with electric times magnetic order equal to 6 in each. The Jordan stabilizer theorem selects the rightmost form.

vector” is a real and duality-invariant statement. It establishes nothing about the carrier: every one of those primitive vectors is exhibited as a sum of two nonzero charge vectors, because label arithmetic always permits decomposition. A bound state of constituents with charges summing to a primitive vector carries the primitive label. Primitive label, divisible carrier.

Second shortcut: simplicity. A simple object of the sector category, one with no internal direct-sum structure, can still be a composite’s channel. The computation is three lines of exact representation arithmetic: in $SU(3)$, the product of two fundamentals contains no singlet, and the product of three contains exactly one, the decomposition $3 \times 3 \times 3$ equal to 1 plus 8 plus 8 plus 10 verified by weight-multiset arithmetic with the dimension check summing to 27 (Figure 3, middle). That unique singlet channel is the proton: a simple, irreducible, color-neutral object that is three quarks. Simplicity certifies the absence of internal sectors, never the absence of constituents.

Third shortcut, and the one this paper adds to the argument: tensor-primality. Say an object is tensor-prime if it factors as a product only trivially. This sounds like the Greek criterion, and the computation shows it is empty in exactly the place it would be needed. In the fusion ring of $SU(2)$, no irreducible above the trivial one factors as a tensor product of smaller nontrivial irreducibles, verified exhaustively through dimension 8: every irreducible is tensor-prime. And every spin above one half arises as a summand of the product of smaller spins, spin one inside half times half, verified for all of them (Figure 3, right). Both facts hold at once because the bound-state relation is a summand of a tensor product, never a factor of one. A composite lives inside A tensor B as a channel; it is never equal to A tensor B. Tensor-primality therefore tests the wrong operation, and any atomicity worth the name must quantify over bound-state constructions, condensations, and confinement functors, the maps that carry constituents to channels, rather than over factorizations.

The hierarchy, the electron object, and the conjecture

The computed no-gos force the definitions upward, and the hierarchy has 7 levels. Label-primitive: the charge vector is primitive, established for the electron and worth nothing alone. Irreducible: the Wigner condition, shared with the proton. Sector-simple: trivial endomorphisms in the sector

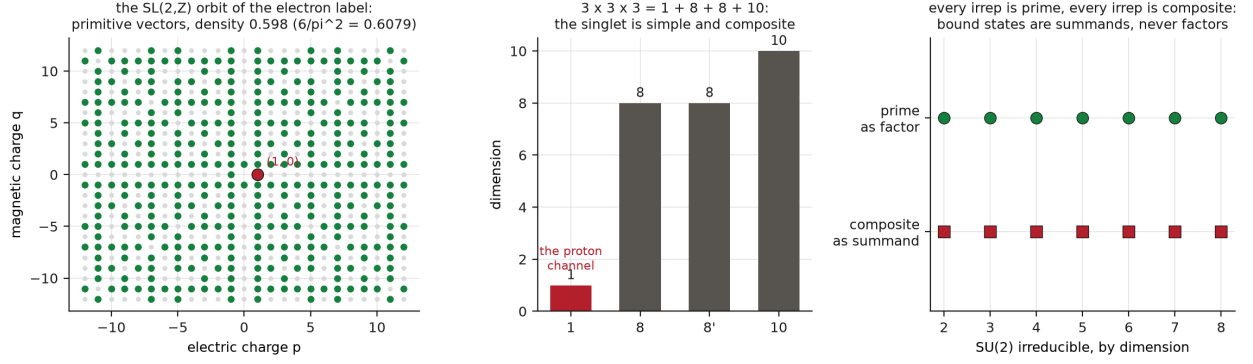


Figure 3: Three shortcuts fail. Left: the $SL(2, \mathbb{Z})$ orbit of the unit electric charge is the primitive vectors, each of which still decomposes into nonzero constituents. Middle: the color singlet is simple and composite: the proton channel in $3 \times 3 \times 3$. Right: in $SU(2)$ fusion, every irreducible is tensor-prime and every irreducible above spin one half is a summand of smaller spins: bound states are summands, never factors.

category. Fusion-prime: no tensor factorization, shown above to be generically vacuous. Bound-state atomic: outside the essential image of every admissible bound-state, condensation, or confinement construction internal to the theory, which is the first level that would have excluded the proton, since the proton is precisely in the image of such a construction applied to quarks. Duality-atomic: bound-state atomic in every exact presentation of M , so that no dual description exhibits the object as a composite, which is the first level that survives the Seiberg obstruction. UV-atomic: duality-atomic in every admissible ultraviolet completion, the closest mathematics comes to the Greek word.

The object to which these predicates should apply is assembled rather than given. Before electroweak symmetry breaking there is no electron; there is a lepton doublet and a charged singlet in different chiral representations. The Yukawa coupling through the Higgs vacuum binds them into one massive Dirac sector, which the paper writes schematically as a cone over the vacuum-mediated map between the two chiral objects, a notation for the assembly rather than a claim of a completed construction. Then the infrared: the massless photon dresses every charged state, and the theorem that charged sectors carry no sharp mass shell (Fröhlich, Morchio and Strocchi, 1979) means the physical object is the dressed sector, defined through infrared regulators and their controlled removal. The electron conjecture, the programme's endpoint, is then a single statement: the dressed post-electroweak electron sector is bound-state atomic in M and remains so under every exact duality of M . Nothing in this paper proves it. The paper's contribution is that the statement now has every term defined, a hierarchy locating it, computed counterexamples showing why each weaker statement would be empty, and a named loophole: should some completion of M present the dressed electron as another sector's bound channel, "elementary" was a fact about our presentation, and the discovery would be larger than the disappointment.

Conjectures and kill criteria

The programme in seven conjectures, compressed to their content. Reconstruction: the exceptional kernel is an isolated point among admissible kernels, with the axioms barred from smuggling the target numbers. Higgs: the kernel's differential calculus produces exactly one electroweak doublet as a fluctuation of the finite Dirac operator and no unwanted light scalars. Family index: some internal operator has index 3, producing three simultaneous generations with nontrivial mixing rather than three embeddings of one. Global form: quantum consistency selects the Z_6 quotient and its computed line lattice uniquely. Infrared: around a dynamically selected Peirce vacuum, the low-energy theory is the Standard Model plus suppressed corrections. Langlands shadow: the selected lattice is the invariant presented to any protected dual description. Atomos: the dressed electron conjecture above.

The kill criteria are the programme's spine, because exceptional mathematics is uniquely vulnerable to retrospective pattern-matching. If the axioms are found to encode 3, 2, or 16 in disguise, there is no derivation. If many inequivalent flags give equally viable worlds, there is no selection. If the flag must forever be chosen by hand, the Standard Model was inserted at the choice. If the spectral calculus cannot produce the Higgs sector, the kernel is representation theory only. If three generations remain an external multiplicity, the deepest structural number is unexplained. If anomalies cannot be trivialized, there is no quantum theory. If the Langlands component never constrains a defect lattice beyond what this paper already computed, it is decoration. If electron atomicity changes under an exact duality, elementarity was presentation-dependent after all. And if no observable consequence ever follows beyond reproducing known numbers, the whole construction is a beautiful reformulation, which is the polite name for a failure.

Objections

The strongest objection is numerological: exceptional structures are large, the Standard Model is specific, and a determined search through large structures finds specific things. The 2026 results partially answer it, because a stabilizer theorem recovering the exact global quotient, and a Peirce chain recovering all six multiplets with correct hypercharges, are single constructions rather than scans, and the ledger computed here shows how overdetermined the target is, a Diophantine system rigid at five of six sites. What the objection still owns is the flag: until a dynamics selects the tripotents, the construction is a coincidence with excellent taste, and this paper's status tables say so.

A second objection holds that the paper's own no-gos undermine its endpoint: if labels, simplicity, and primality all fail, perhaps atomicity is the wrong concept entirely. The reply is that the failures are the finding: they show exactly which quantifier the concept needs, over bound-state constructions and over presentations, and the DHR tradition shows that quantifying over a sector category is mathematically actionable rather than rhetorical (Doplicher and Roberts, 1989).

A third objection: the Langlands material is a famous name attached to a speculative paper. The paper's use of it is deliberately minimal, one computed lattice and one disciplined conjecture, with the

supersymmetric provenance of the rigorous results stated every time they are mentioned. Where the shadow cannot be computed, it is not invoked.

A fourth objection: nothing here is a physical prediction. Almost: the global form is one. The four theories of Figure 2 differ in their monopole and dyon spectra, and a kernel that selects the Z_6 quotient commits to one column of allowed magnetic charges. That is a falsifiable structural commitment, far from a cross-section, and stated at its true size.

Falsification

The programme fails as specified by its kill criteria, and the paper's own contributions fail on narrower grounds. The ledger computation fails if any of the five sums is nonzero under the stated conventions, or if some deletion other than the right-handed neutrino leaves all conditions intact, since the rigidity claim is exact. The global-form atlas fails if the six matter classes do not generate a Z_6 , if some multiplet is inadmissible under some form, or if the four lattices are not distinct, each a finite check reproduced by the simulation. The $SL(2, Z)$ theorem is proved, so what could fail is its relevance: a demonstration that duality-primitive labels do imply carrier indivisibility in some class of theories would overturn the no-go and simplify the hierarchy. The summand-never-factor lesson fails if a physical bound-state construction is exhibited whose output is a tensor factor rather than a channel, which would rehabilitate fusion-primality. And the programme's central bet, that the exceptional kernel is isolated among admissible kernels, is exactly the kind of claim a classification computation could kill in an afternoon, which is why the paper states it as a conjecture and specifies the moduli problem.

The last loophole

Suppose everything succeeds. The kernel classified and isolated, the flag selected by dynamics, the Higgs produced, the families indexed, the quantum completion built, the dualities quotiented, and the dressed electron proved bound-state atomic in every presentation of M . The theorem would say: within M , there is nothing for the electron to factor into. It would not say that reality has no deeper theory beneath M , in which the prime object of M is the infrared shadow of something composite, as the effective proton is elementary in its own effective theory and three quarks in the ultraviolet. Primality is relative to the category in which composition is defined. The only escape is a uniqueness theorem about M itself, that the complete observable structure admits no inequivalent completion, and nothing in mathematics or physics currently approaches such a statement. The honest endpoint of the programme is therefore conditional in a way Democritus would recognize: *atomos* was never a discovery about objects, it was a commitment about explanation, the claim that cutting must stop somewhere. What the mathematics of 2026 has changed is the quality of the question. It is now possible to say, in computable terms, what stopping would look like: a distinguished object, in a uniquely reconstructed category, outside the image of every bound-state functor, invariant under every duality, carrying the one charge lattice the kernel selects. The electron is not yet that object. For the first time, the sentence has a truth value to await.

Reproducibility

The simulation, its 3 figures, and every model number quoted above regenerate from one command in the paper’s repository (`simulation/run_all.py`). Everything is exact integer and rational arithmetic, bit for bit on rerun. 17 invariants, among them the five consistency sums vanishing identically, the deletion-rigidity count with the right-handed neutrino as the unique free site, the Z_6 screening subgroup generated by the matter classes, the admissibility of all six multiplets under all four global forms with electric and magnetic orders multiplying to 6 and the Dirac pairing integral, the constructive coprimality of the $SL(2, \mathbb{Z})$ orbit with the primitive density within 2 percent of 6 over π squared, the unique singlet in the triple product of color fundamentals with its dimension check, and the universal tensor-primality alongside universal summand-compositeness in the $SU(2)$ fusion ring, fail the run loudly if broken.

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